

I would investigate few-label industrial defect inspection with a photonic reservoir

A measurable 12-week path from public data to a conditional hardware test

NGUYEN NGO NHAT NAM · R&D QUANTUM OPTICS ENGINEER CANDIDATE · 15 SEP 2026 · 15:30 GMT+7

REAL INDUSTRIAL NEED

Scarce expert labels

Quality inspection needs useful accuracy from small labelled sets.

PHOTONIC FIT

Fixed nonlinear map

Interference transforms compact phase-encoded image features.

12-WEEK TEST

Decision-ready result

Public data · matched controls simulator → conditional QPU run

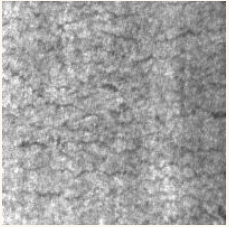
COMPLETED DE-RISKING PILOT

Q1 0.8708 macro-F1 · 48 labels per class

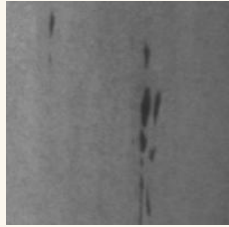
Useful signal; no advantage yet

Evidence informs the next test; it does not support a quantum-advantage claim.

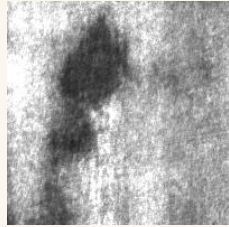
Why this is a good 12-week feasibility study



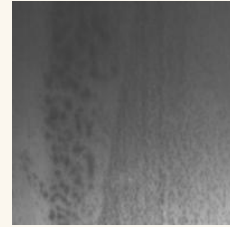
Crazing



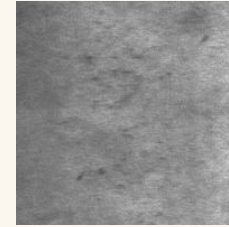
Inclusion



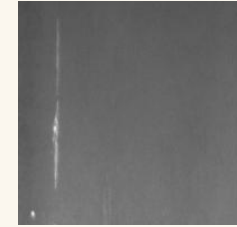
Patches



Pitted surface



Rolled-in scale



Scratches

REAL NEED

Labels are expensive

The target is useful accuracy from
12 · 24 · 48 labels per class.

Sample efficiency is measurable.

TRACTABLE BENCHMARK

1,800 images

Six balanced defect classes
300 images per class

Public data · macro-F1 target

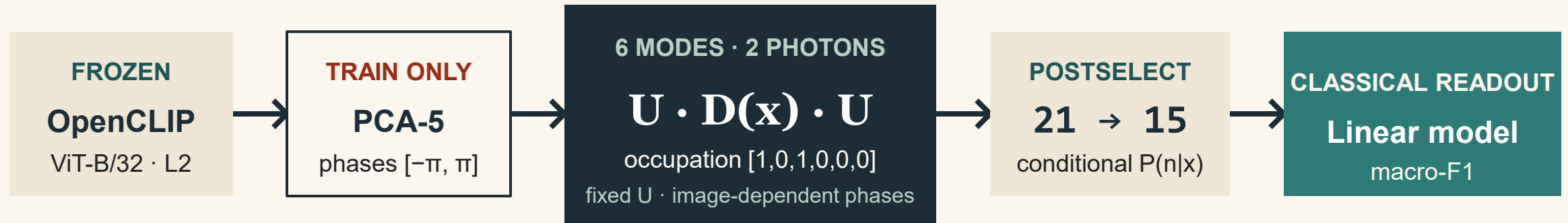
LEAKAGE-SAFE EVIDENCE

One duplicate group

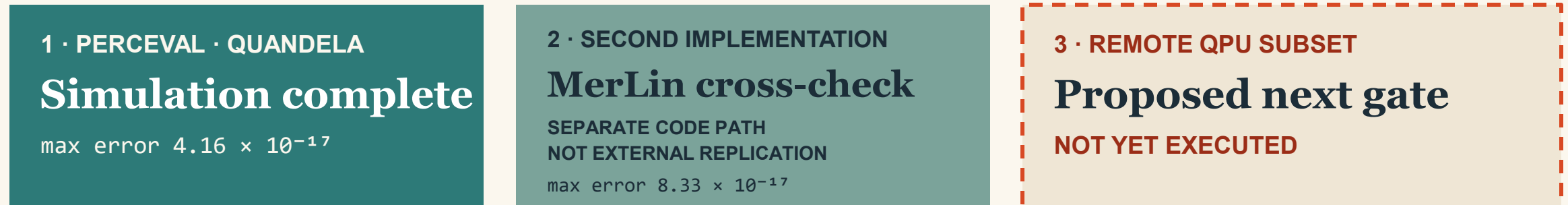
Pa_101 · Pa_105 never cross splits.
Fold 0: 60 test images per class.

Real need · data · decision metric

A small reservoir fits Quandela's path to hardware



$$U(x) = UD(x)U, \quad \phi_Q(x) = \text{Normalize}(\{P(\mathbf{n} | x)\}_{\mathbf{n} \in \mathcal{C}})$$



The feasibility question is deliberately falsifiable

AT THE SAME PCA-5 INPUT

Can Q1's 15-D photonic features beat matched 15-D controls and approach PCA-5 RBF-SVM?

PAIRED PROTOCOL

12 · 24 · 48

labels per class

Subset seeds 11 · 22 · 33

Map seeds 101 · 202 · 303

Validation macro-F1 only

One test evaluation per

selected fitted pipeline

MAP SEEDS AVERAGED · NEVER PICKED

7 FAMILIES · EQUAL-WIDTH SCHEMATIC

B1 Full CLIP + linear

B2 Full CLIP + RBF-SVM

B3 PCA-5 + linear

B4 PCA-5 + RBF-SVM

B5 RFF-15

B6 ELM-15

Q1 Photonic-15

PASS A · ACCURACY PARITY

$Q1@48 \geq B4@48 - 0.02$

on at least 4 / 5 outer folds

PASS B · 2× LABEL EFFICIENCY

$Q1@24 \geq B4@48 - 0.02$

≥4 / 5 folds · ≤2,000 shots / image
acceptance rate ≥ 0.70

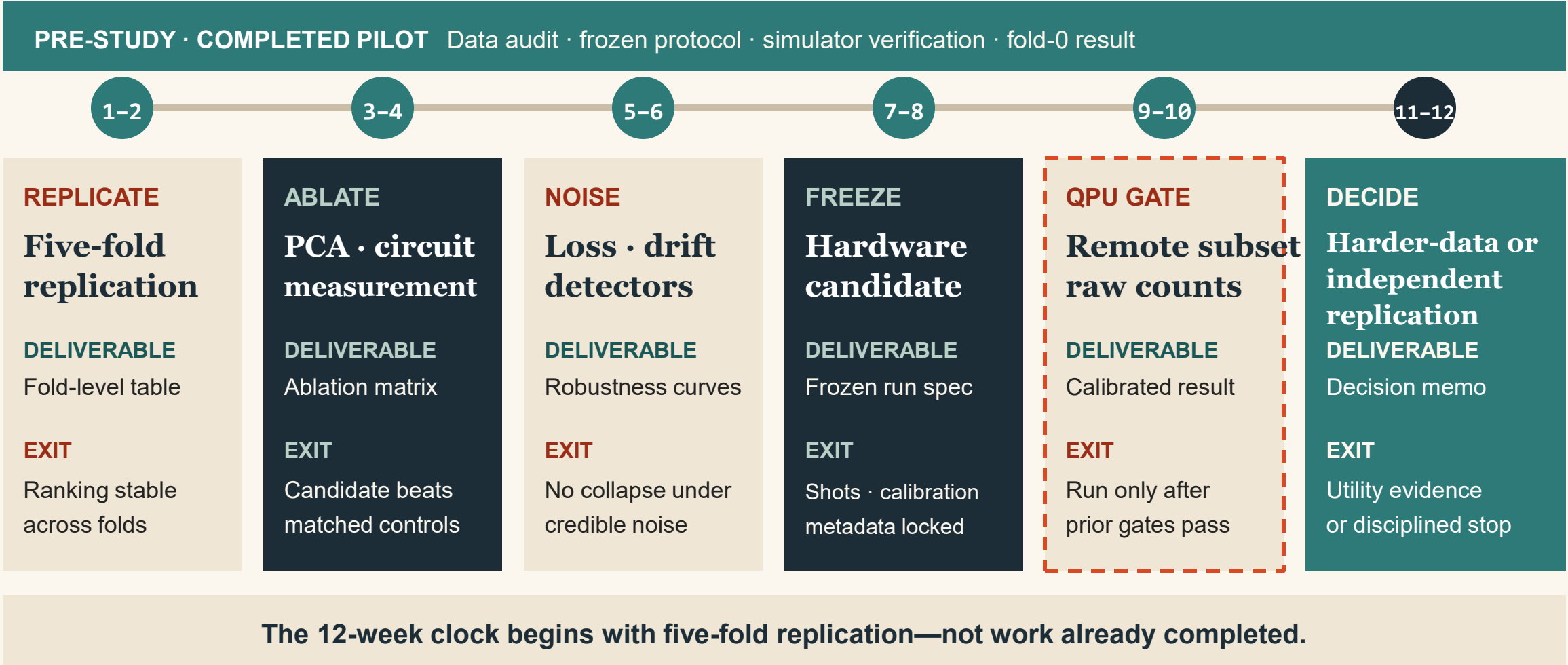
STOP IF NEITHER PASS HOLDS

or Q1 loses to RFF-15 and ELM-15 on ≥4 / 5 folds

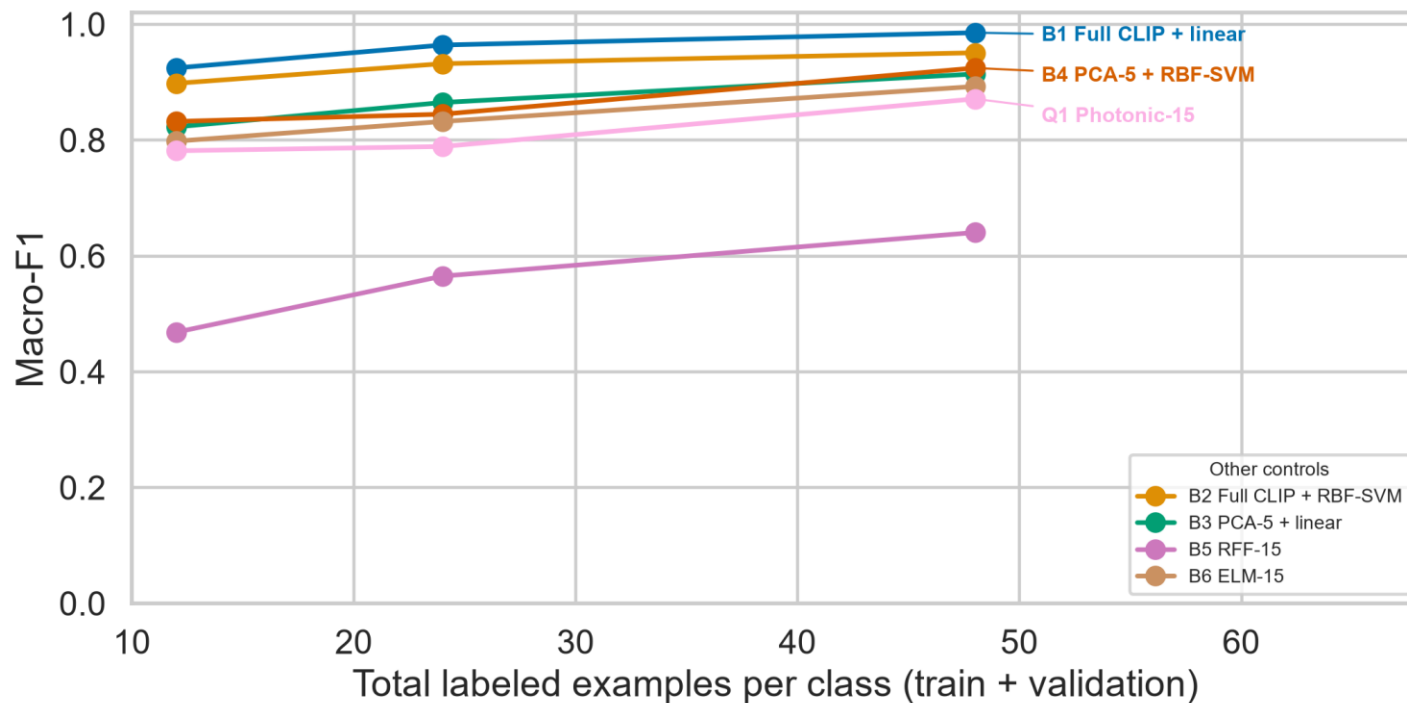
Matched 15-D controls: Random Fourier Features (RFF) and Extreme Learning Machine (ELM).

B4 = PCA-5 radial-basis-function support vector machine (RBF-SVM); resource value = label efficiency under shot and acceptance constraints.

The 12-week project starts after the completed pilot



The pilot shows useful signal—not competitive advantage



Q1 · PHOTONIC-15

0.7812 → 0.7886

→ 0.8708

12 · 24 · 48 labels per class

B1 · FULL CLIP + LINEAR

0.9242 → 0.9638

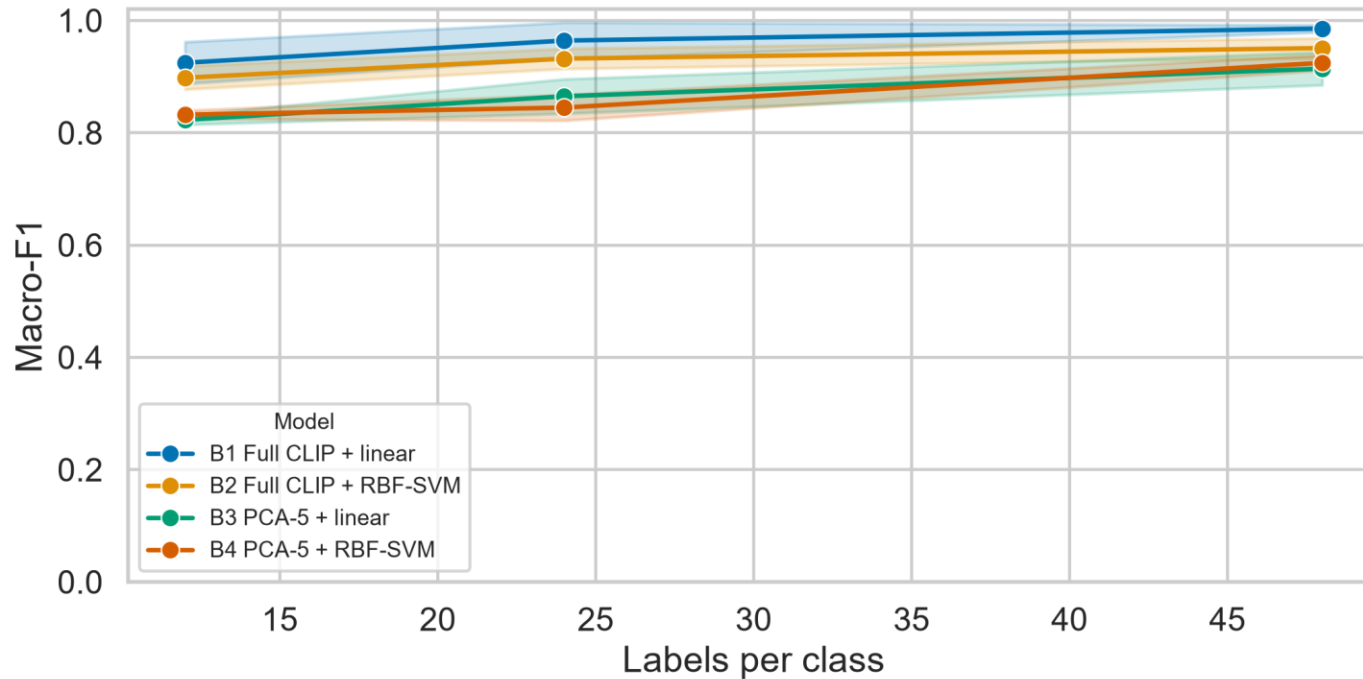
→ 0.9852

PILOT VERDICT

Learns signal;
not competitive yet.

Mean ± sample SD across three subset seeds · one outer fold · source: main_results.csv

PCA-5 explains most of the gap; the map explains the rest



FULL → PCA-5 PENALTY · B1 – B3

+.1018 · +.0993 · +.0713

budgets 12 · 24 · 48

Q1 – RFF

+.3133 · +.2238 · +.2307

Q1 – ELM

–.0166 · –.0432 · –.0215

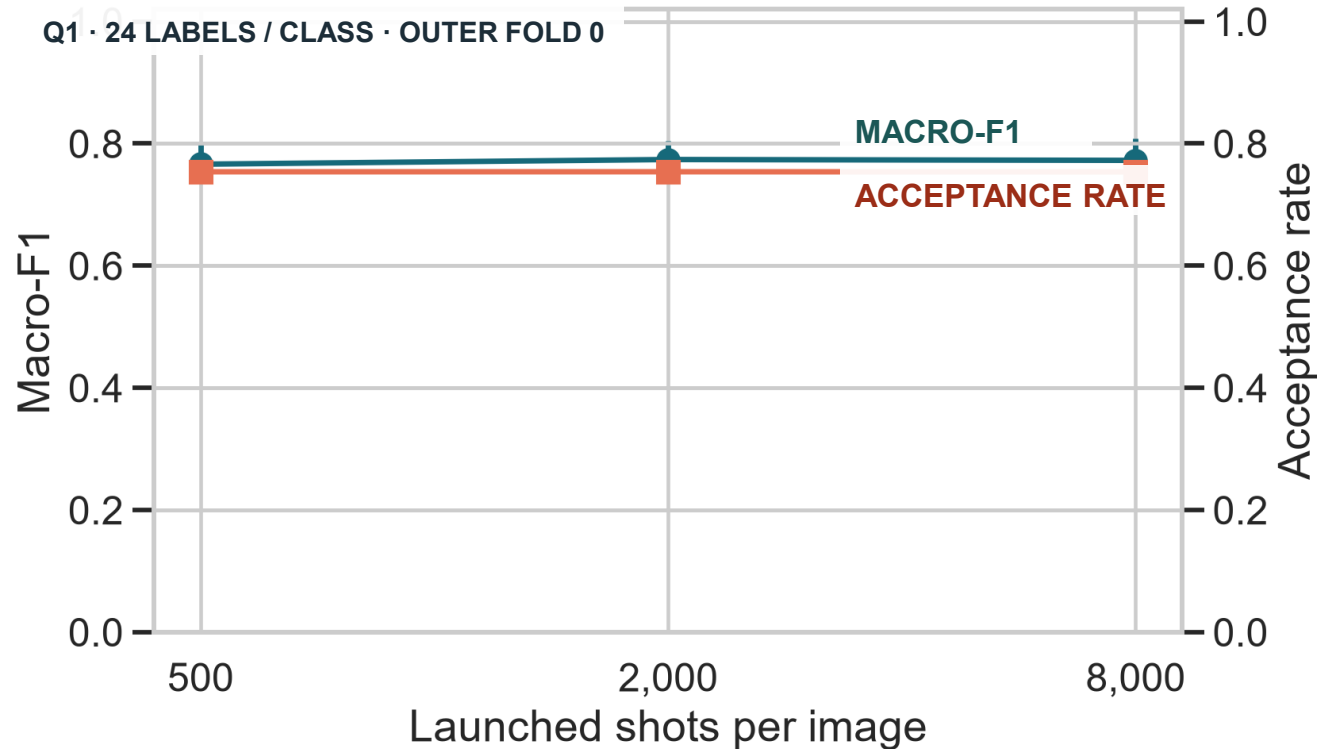
Q1 – PCA-5 RBF-SVM

–.0507 · –.0558 · –.0534

Only win: locked matched RFF-15 comparator.

Separate information lost in PCA-5 from the value added by the photonic circuit.

Finite-shot sampling is stable; accuracy does not recover



MACRO-F1 · 15 EVALUATIONS EACH

500 $0.7662 \pm .0322$

2,000 $0.7739 \pm .0308$

8,000 $0.7725 \pm .0363$

acceptance ≈ 0.754

0 zero-acceptance failures

TESTED

Shot sampling
Acceptance
Ideal-trained readout

NOT YET TESTED

Loss · phase drift
Distinguishability
Detectors · noise training

45 evaluations = 3 map seeds × 5 sampling seeds × 3 shot levels · same outer test fold · source: finite_shot.csv

Simulation pipeline demonstrated; hardware feasibility untested

DEMONSTRATED

Simulation pipeline

Executable pipeline
Simulator consistency
Reproducible artifacts

PARTIALLY ANSWERED

Application signal

Learning with labels
Paired one-fold ranking
Finite-shot sampling

UNANSWERED

Practical utility

Five-fold generalization
Realistic photonic noise
Remote hardware execution

OPENCLIP EMBEDDINGS · CPU

96.57 s

One cached feature pass
Reused across every model family.

PILOT MODEL GRID · CPU

≈59.9 s

Broader validation is tractable
Five folds fit the project window.

DECISION CONSTRAINT

Scientific validity

One outer fold · small validation cells
ideal optics · B1 near ceiling
RUNTIME IS NOT THE BOTTLENECK

Local CPU profile · no loss, phase drift, detector noise, or QPU queue

Continue only through a decision-changing test

WHICH APPLICATION?

Few-label industrial defect inspection

with a fixed photonic reservoir

WHY THIS IDEA?

Real label scarcity · compact phase encoding

Quandela simulator → hardware path · 12-week evidence

CURRENT VERDICT

Feasible in simulation and scientifically interesting

HARDWARE FEASIBILITY REMAINS UNTESTED

CONTINUE ONLY IF · B4 = PCA-5 RBF-SVM

PASS A · $Q1@48 \geq B4@48 - 0.02$

↳ on ≥ 4 / 5 folds

PASS B · $Q1@24 \geq B4@48 - 0.02$

↳ on ≥ 4 / 5 folds · $\leq 2,000$ shots / image

↳ acceptance ≥ 0.70

STOP CONDITION

Neither pass holds

or Q1 loses to RFF-15 and ELM-15 on ≥ 4 / 5 folds.

Evidence first · hardware only after gates pass

Repository · github.com/MonsieurNam/Quanova_Photonic_Defect_QML